

# Refutation of conjecture 00000001671: the basis number of $K_n$ is *not* $\lceil (n+3)/2 \rceil$

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## Abstract

Conjecture 00000001671 asserts that the basis number of the complete graph  $K_n$  equals  $\lceil (n+3)/2 \rceil$ . We show this is false. The true values for  $n = 3, 4, 5, 6$  are 1, 2, 3, 3 respectively, whereas the formula predicts 3, 4, 4, 5. All four cases are verified by exact computation; the cases  $n = 3$  and  $n = 4$  are additionally verified in core Lean 4 with no axioms. We also observe that the claim fails under the alternative reading in which “basis number” means the edge multiplicity of a fixed canonical basis.

## 1 Definitions and the claim

Throughout,  $G = (V, E)$  is a finite connected graph and all linear algebra is over the field  $\mathbb{F}_2$  with two elements. We identify an edge subset  $S \subseteq E$  with its characteristic vector in  $\mathbb{F}_2^E$ ; the *cycle space*  $\mathcal{C}(G) \subseteq \mathbb{F}_2^E$  is the kernel of the incidence map, i.e. the set of edge subsets in which every vertex has even degree. It is well known that  $\dim \mathcal{C}(G) = |E| - |V| + 1$  when  $G$  is connected, and that an edge subset is a cycle in this sense exactly when it is a disjoint union of (edge sets of) cycles of  $G$ .

**Definition 1** (cycle basis). A *cycle basis* of  $G$  is a set  $\mathcal{B} \subseteq \mathcal{C}(G)$  of linearly independent elements that spans  $\mathcal{C}(G)$ ; equivalently,  $|\mathcal{B}| = \dim \mathcal{C}(G)$  and  $\text{span } \mathcal{B} = \mathcal{C}(G)$ .

**Definition 2** ( $k$ -fold basis; basis number). A cycle basis  $\mathcal{B}$  is  *$k$ -fold* if every edge of  $G$  lies in at most  $k$  of the cycles in  $\mathcal{B}$ . The *basis number*  $b(G)$  is the least  $k$  for which a  $k$ -fold cycle basis exists.

**Definition 3** (the conjecture). Conjecture 00000001671 states that

$$b(K_n) = \lceil \frac{n+3}{2} \rceil \quad (n \geq 3). \tag{1}$$

**Theorem 4** (Verdict). *The claim (1) is false. The true basis numbers are*

|                         |   |   |   |   |
|-------------------------|---|---|---|---|
| $n$                     | 3 | 4 | 5 | 6 |
| $b(K_n)$                | 1 | 2 | 3 | 3 |
| $\lceil (n+3)/2 \rceil$ | 3 | 4 | 4 | 5 |

(2)

*In particular (1) fails for every  $n \in \{3, 4, 5, 6\}$ .*

The remainder of the paper proves (2) case by case and then discusses the strongest alternative reading of the conjecture.

*Remark 5* (two elementary counting bounds). Let  $d = \dim \mathcal{C}(G) = |E| - |V| + 1$  and let  $\mathcal{B}$  be a  $k$ -fold cycle basis.

- (i) Every cycle of a simple graph has at least 3 edges, so  $3d \leq \sum_{C \in \mathcal{B}} |C| \leq k|E|$  and hence  $k \geq \lceil 3d/|E| \rceil$ .
- (ii) If  $G$  is simple, the sum  $\sum_{C \in \mathcal{B}} |C|$  counts each edge with multiplicity at most  $k$ ; when equality  $\sum_{C \in \mathcal{B}} |C| = k|E|$  holds, *every* edge is used *exactly*  $k$  times.

## 2 $K_3$ : the basis number is 1, not 3

Here  $|V| = 3$ ,  $|E| = 3$ , so  $d = 3 - 3 + 1 = 1$ . The cycle space consists of  $\emptyset$  and the single triangle  $C = \{01, 02, 12\}$ ; this is the unique nonzero cycle, and  $\{C\}$  is a cycle basis. Each of the three edges lies in exactly one basis cycle, so  $\{C\}$  is 1-fold and  $b(K_3) \leq 1$ . The value 0 is impossible because a 0-fold basis would have to consist of cycles using no edge at all, which cannot span the nonzero cycle space. Hence

$$b(K_3) = 1 \neq 3 = \lceil \frac{3+3}{2} \rceil.$$

This already refutes (1). It also shows that the assertion in the conjecture that “the known lower bound matches” is inaccurate: the elementary lower bound of Remark 5(i) is  $\lceil 3 \cdot 1/3 \rceil = 1$ , and it is tight here, not equal to 3.

## 3 $K_4$ : the basis number is 2, not 4

Now  $|V| = 4$ ,  $|E| = 6$ , so  $d = 6 - 4 + 1 = 3$ .

**No 1-fold basis.** A 1-fold basis  $\mathcal{B}$  of three independent cycles would satisfy

$$9 = 3d \leq \sum_{C \in \mathcal{B}} |C| \leq 1 \cdot |E| = 6,$$

an impossibility.

**A 2-fold basis.** Consider the three triangles (written as vertex sets)

$$T_1 = \{0, 1, 2\}, \quad T_2 = \{0, 1, 3\}, \quad T_3 = \{0, 2, 3\}.$$

Their edge sets are

$$E(T_1) = \{01, 02, 12\}, \quad E(T_2) = \{01, 03, 13\}, \quad E(T_3) = \{02, 03, 23\}.$$

They are linearly independent over  $\mathbb{F}_2$ : the edge 12 lies only in  $T_1$ , the edge 13 only in  $T_2$ , and the edge 23 only in  $T_3$ , so any linear relation has all coefficients zero. Since  $d = 3$ , they form a basis. Edge multiplicities are:

| edge                           | 01 | 02 | 03 | 12 | 13 | 23 |
|--------------------------------|----|----|----|----|----|----|
| # {basis cycles containing it} | 2  | 2  | 2  | 1  | 1  | 1  |

Hence every edge lies in *at most* two of the basis cycles, so this is a 2-fold basis and  $b(K_4) \leq 2$ . Combined with the previous paragraph,  $b(K_4) = 2$ .

*Remark 6* (a caution about the witness). It is *not* true that every edge of  $K_4$  lies in exactly two of the three triangles  $T_1, T_2, T_3$ : the edges 12, 13, 23 of the omitted triangle  $\{1, 2, 3\}$  lie in only one of them. The correct statement is that every edge lies in *at most* two, which is precisely what is needed for the value  $k = 2$ . (If instead one took all four triangles, edge 01 would lie in three of them, and the set would not even be independent.)

Therefore

$$b(K_4) = 2 \neq 4 = \lceil \frac{4+3}{2} \rceil.$$

#### 4 $K_5$ : the basis number is 3, not 4

Here  $|V| = 5$ ,  $|E| = 10$ , so  $d = 10 - 5 + 1 = 6$ .

**No 1-fold or 2-fold basis.** For  $k = 1$ , Remark 5(i) gives  $k \geq \lceil 3 \cdot 6/10 \rceil = 2$ , so no 1-fold basis exists.

For  $k = 2$  the total-length budget is  $k|E| = 20$ , while  $3d = 18$ ; so

$$\sum_{C \in \mathcal{B}} (|C| - 3) \leq 20 - 18 = 2.$$

Every cycle has length at least 3, and a cycle of length  $\geq 6$  would already force the sum to be at least  $6 + 5 \cdot 3 = 21 > 20$ . Hence a hypothetical 2-fold basis would consist of cycles of lengths 3, 4, 5 only, with at most two units of “excess”  $\sum (|C| - 3)$ . Since  $K_5$  has  $\binom{5}{3} = 10$  triangles,  $\binom{5}{4} \cdot 3 = 15$  four-cycles, and  $\binom{5}{5} \cdot (5 - 1)!/2 = 12$  five-cycles, the number of candidate 6-element cycle sets of total length at most 20 is exactly

$$\underbrace{\binom{10}{6}}_{210} + \underbrace{\binom{10}{5} \cdot 15}_{3780} + \underbrace{\binom{10}{5} \cdot 12}_{3024} + \underbrace{\binom{10}{4} \binom{15}{2}}_{22050} = 29\,064.$$

Exhaustive check of these 29 064 sets shows that none is linearly independent with every edge occurring at most twice. (The check is reproduced by the accompanying `reproduce.py`; the same run also performs a brute-force check of all bases for  $K_3$  and  $K_4$ .) Thus no 2-fold basis exists.

**A 3-fold basis.** Fix the vertex 0 as a centre and take, for each pair  $1 \leq i < j \leq 4$ , the triangle  $C_{ij} = \{0, i, j\}$ . There are  $\binom{4}{2} = 6$  of them, equal to  $d$ , and they are independent (the edge  $ij$  occurs only in  $C_{ij}$ ). The spokes  $0i$  occur in the three triangles  $C_{ij}$  with  $j \neq i$ , and the edges  $ij$  occur once; hence every edge occurs in at most 3 cycles. So  $b(K_5) \leq 3$ , and with the previous paragraph,

$$b(K_5) = 3 \neq 4 = \lceil \frac{5+3}{2} \rceil.$$

#### 5 $K_6$ : the basis number is 3, not 5

Here  $|V| = 6$ ,  $|E| = 15$ , so  $d = 15 - 6 + 1 = 10$ .

**No 1-fold or 2-fold basis.** For  $k = 1$ , Remark 5(i) gives  $k \geq \lceil 3 \cdot 10/15 \rceil = 2$ , so no 1-fold basis exists.

For  $k = 2$ , the budget is  $k|E| = 30$  and  $3d = 30$  are equal. By Remark 5(ii), a 2-fold basis would consist of 10 triangles in which *every* edge of  $K_6$  occurs exactly twice. Such a family is a Steiner triple system on 6 points: a set of triples in which every unordered pair of points lies in exactly one triple. But in a Steiner triple system on  $v$  points each point lies in

$$\frac{v-1}{2} = \frac{6-1}{2} = \frac{5}{2}$$

triples, which is not an integer. Contradiction. Hence no 2-fold basis exists.

**A 3-fold basis.** The following ten triangles (vertex sets) of  $K_6$  form a 3-fold basis:

$$\{0, 1, 2\}, \{0, 1, 3\}, \{0, 1, 4\}, \{0, 2, 3\}, \{0, 2, 4\}, \\ \{0, 3, 5\}, \{0, 4, 5\}, \{1, 2, 5\}, \{1, 3, 4\}, \{1, 3, 5\}.$$

They are linearly independent (their span has dimension  $10 = d$ ), and a direct count gives the edge multiplicities

| edge  | 01 | 02 | 03 | 04 | 05 | 12 | 13 | 14 | 15 | 23 | 24 | 25 | 34 | 35 | 45 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| mult. | 3  | 3  | 3  | 3  | 2  | 2  | 3  | 2  | 2  | 1  | 1  | 1  | 1  | 2  | 1  |

so every edge occurs at most three times. Therefore  $b(K_6) \leq 3$ , and with the previous paragraph,

$$b(K_6) = 3 \neq 5 = \lceil \frac{6+3}{2} \rceil.$$

This completes the proof of Theorem 4.

## 6 The strongest objection: a fixed canonical basis

A possible defence of the conjecture is to change the meaning of “basis number” from Definition 2 to the maximum edge multiplicity of some *specific* canonical cycle basis, for instance the standard *star basis* of  $K_n$ .

Fix a vertex 0 and let  $T$  be the star with centre 0, i.e. the spanning tree with edges  $0i$  for  $1 \leq i \leq n-1$ . The non-tree edges are the pairs  $ij$  with  $1 \leq i < j \leq n-1$ , of which there are  $\binom{n-1}{2} = d$ . The corresponding fundamental cycles are the triangles  $0-i-j-0$ , and they form the star basis. In this basis the spoke  $0i$  lies in one triangle for every  $j \in \{1, \dots, n-1\} \setminus \{i\}$ , i.e. in exactly  $n-2$  basis cycles, while every edge  $ij$  lies in exactly one. The maximum edge multiplicity of the star basis is therefore

$$n-2, \tag{3}$$

and the star basis is optimal among fundamental bases of a star.

Under this alternative reading, the predicted value for  $K_n$  would be  $n-2$ ; for  $n = 4, 5, 6$  this is 2, 3, 4. The conjectured formula  $\lceil (n+3)/2 \rceil$  gives 4, 4, 5 for the same values of  $n$ . Hence

$$n-2 \neq \lceil \frac{n+3}{2} \rceil \quad (n = 4, 5, 6), \tag{4}$$

so the formula is wrong under this reading as well. In other words, the claim is false both for the least- $k$  definition (Definition 2, which is the definition stated in the conjecture) and for the fixed-canonical-basis reading. The two readings only happen to agree at  $n = 3$ , where both give 1, and even there the formula  $\lceil (3+3)/2 \rceil = 3$  is wrong.

*Remark 7.* The star basis does give the optimal value  $b(K_4) = 2$  and  $b(K_5) = 3$ , which is presumably the source of the conjecture’s “upper bound”. It fails to be optimal already for  $K_6$ , where  $n-2 = 4$  while the true value is 3 (Section 6); the ten-triangle basis above is strictly better than the star basis.

## 7 Machine verification

All statements of Theorem 4 are verified by exact computation.

- `reproduce.py` (Python standard library only) enumerates the cycle space of  $K_n$  over  $\mathbb{F}_2$ , and decides the existence of a  $k$ -fold basis by an exact backtracking search using the length budget of Remark 5. It prints the table (2) against the formula and exits 0 on **PASS**. For  $K_3, K_4$  it independently re-derives the values by exhaustive enumeration of *all* bases.
- `lean4/` contains a core-Lean 4 development (no Mathlib, no `sorry`, no `axiom`, no `native_decide`) in which edge subsets are bitmasks and the cycle space, the predicate `IsKFoldBasis`, and `minBasisNumber` are straight structural recursions, so that the kernel itself evaluates them. It proves `minBasisNumber 3 = 1` and `minBasisNumber 4 = 2` by `decide`, together with the mismatches against  $\lceil (n+3)/2 \rceil$ . The axiom audit reports that none of these theorems depends on any axiom.